Aayush Bhat

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Portfolio: aayush-portfolio-390a.onrender.com

Summary

Backend and data engineer with production experience in distributed backend systems, RESTful APIs, and event-driven pipelines (Python, Java, SQL); ships Spark/Parquet platforms on AWS and Azure.

Education

University School of Information, Communication and Technology — Delhi, India
Master of Computer Applications, Software Engineering; GPA: ***8.9/10*** — Aug 2024 – May 2026

Integrated Institute of Technology — Delhi, India
Bachelor of Computer Applications; GPA: ***9.01/10*** — Aug 2020 – May 2023

Skills

Languages & Databases: C++, Python, Java, SQL, TypeScript, JavaScript, PostgreSQL, MongoDB, Redis, SQLite, non-relational databases, T-SQL
Web Technologies: Next.js, FastAPI, Node.js, Express.js, React, Socket.IO, REST APIs, GraphQL, HTML, CSS, Tailwind CSS, Three.js, Prisma, NextAuth
Cloud & Platforms: Azure Data Factory, Databricks, Apache Spark (PySpark), Delta Lake, Unity Catalog, DuckDB, AWS, AWS Lambda, Amazon EMR, OCI, Hugging Face, Render, Neon, Azure SQL, Power BI
Core Concepts: Data Structures & Algorithms (DSA), Data Modeling, Data Pipelines, Streaming Systems, Columnar Data Models, Microservices, ETL, Hadoop, Hive, NLP, MinHash, TF-IDF, Parquet, PyArrow, OAuth 2.0, OpenID Connect, JWT, RBAC, Argon2id, TOTP, CSRF, Distributed Systems, Data Warehousing, Data Governance, Medallion Architecture, Data Quality
Testing, Tools & Observability: Docker, GitHub Actions, CI/CD, Azure Monitor, Application Insights, Git, pytest

Experience

Nagarro — Gurugram, India
Associate Software Engineer — Sep 2026 – Present
Associate Software Engineer Trainee — Jan 2026 – Sep 2026

- Built backend microservices and REST APIs using **Python** and **FastAPI**, reducing data-retrieval latency by **65%** with **DuckDB** zero-copy OLAP queries over **SQL** data.
- Processed **3.8M+** daily streaming telemetry events with **PySpark** and built window-function validation across **150+** streams, achieving **99.9%** data accuracy.
- Managed production deployments and **GitHub Actions** CI/CD; used **Azure Monitor** to cut incident-resolution time by **30%** and automated batch jobs across **85+** tables, improving deployment speed by **40%**.
- Co-built an **Azure Data Factory** and **Azure SQL** medallion POC for simulated EV-charger telemetry (**25K** rows/day), with **8** pipelines, a **12**-rule exception layer and packet-gap detection, **15** data-quality checks, and a **6**-page **Power BI** report.
- Co-built a **PySpark** and **Delta Lake** medallion POC for simulated HVAC telemetry (**81** chillers, **140K** readings/day), adding rule-based data quality with quarantine and watermark-driven incremental runs across **24** Gold tables.

Projects

Globe News Atlas | *Live Demo*
Python, FastAPI, React, TypeScript, SQL, Parquet, PyArrow, SQLite, Three.js, GitHub Actions, Docker, pytest, OAuth 2.0, JWT

- Built a full-stack world-news atlas ingesting nine public sources into a bronze/silver/gold **Parquet** data lake on **Hugging Face**, serving **400-event** 24-hour snapshots through a **FastAPI** API and a React/Three.js WebGL globe.
- Engineered an ETL/NLP pipeline with URL canonicalization, **MinHash LSH** deduplication, **TF-IDF/LSA** clustering, and geocoding, narrowing 123 candidate arcs to 19 evidence-backed storylines.
- Implemented authentication in FastAPI and SQLite using **Argon2id** hashing, CSRF protection, rate limiting, email validation, **TOTP** 2FA, and Google **OAuth 2.0/OpenID Connect** with PKCE and JWT validation.

TaskFlow – Chat-Native Work OS | *Live Demo* | *GitHub*
Next.js 14, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma, Socket.IO, NextAuth, Docker, Render, Neon

- Built and deployed a chat-native work OS spanning tasks, notes, mind maps, calendar, focus sessions, and chat across **62 Prisma models**, as a single **Docker** service on Render with Neon PostgreSQL and startup migrations.
- Engineered real-time collaboration with **Socket.IO** rooms, including threads with live reply counts, reactions, typing indicators, presence, pinned decisions, and read state, backed by REST APIs and Prisma.
- Architected role-based access control (RBAC) with **JWT** across **4** isolated containerized services, securing **50+** REST endpoints and granular ADMIN, EDITOR, and READ_ONLY policies across **100+** workspaces.

Achievements

- LeetCode contest rating of **2021** (Top **1.57%** globally), **Knight** status, and **750+** problems solved.
- Secured All India Rank **1,030** in the NIMCET 2024 national entrance examination among **50,000+** candidates.

Certifications

- **Oracle Cloud Infrastructure 2025 Foundations Associate** | *Credential* (Oct 19, 2025; valid to Oct 19, 2027).
- **Complete React Developer (w/ Redux, Hooks, GraphQL)** | *Credential* (Jul 10, 2025; 42 hours).
- **HackerRank REST API (Intermediate)** | *Credential* (Oct 7, 2025).